

Numerical Computing II - Assignment 3

Name: Saily Kadam

Roll no: 2511

Subject: Numerical Computing II - Assignment 3

Investigation of Richardson Extrapolation

Plot of Functions:

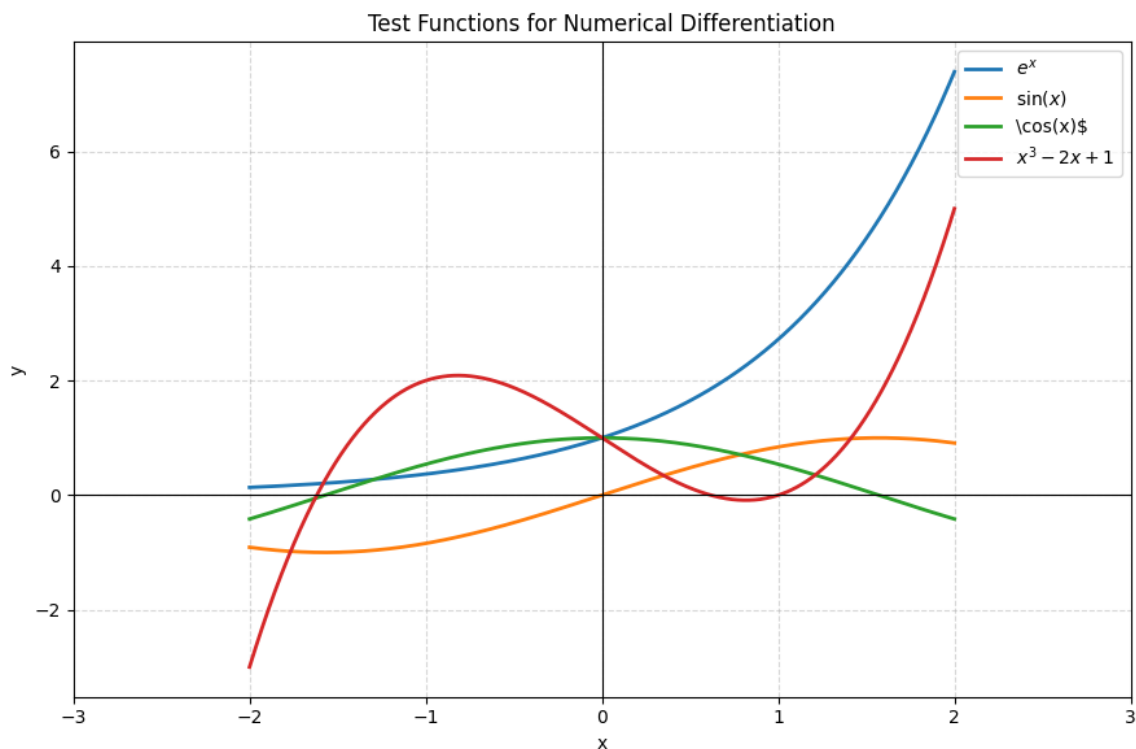


Figure 1: Visual representation of the four test functions

Algorithm / Pseudocode

CentralDifference(f, x, h)

```
{  
    f_x_plus = f(x + h)  
    f_x_minus = f(x - h)  
    derivative = (f_x_plus - f_x_minus) / (2 * h)  
    RETURN derivative
```

```
}
```

```
RichardsonExtrapolation(f, x, h)
```

```
{
```

```
    D_h = CentralDifference(f, x, h)
```

```
    D_h2 = CentralDifference(f, x, h / 2)
```

```
    R = (4 * D_h2 - D_h) / 3
```

```
    RETURN R
```

```
}
```

Results Tables

Function: $f(x) = e^x$ (at $x = 1$)

h	Exact	D_h	Error_D	R_h	Error_R
0.1	2.71828	2.72281	0.00453274	2.71828	5.66477e-07
0.01	2.71828	2.71833	4.53049e-05	2.71828	5.66911e-11
0.001	2.71828	2.71828	4.53047e-07	2.71828	4.02345e-13
0.0001	2.71828	2.71828	4.53057e-09	2.71828	8.5576e-13
1e-05	2.71828	2.71828	5.85869e-11	2.71828	5.85869e-11
1e-06	2.71828	2.71828	1.63457e-10	2.71828	1.32602e-10
1e-07	2.71828	2.71828	5.85874e-11	2.71828	2.90201e-09
1e-08	2.71828	2.71828	6.60275e-09	2.71828	6.60275e-09

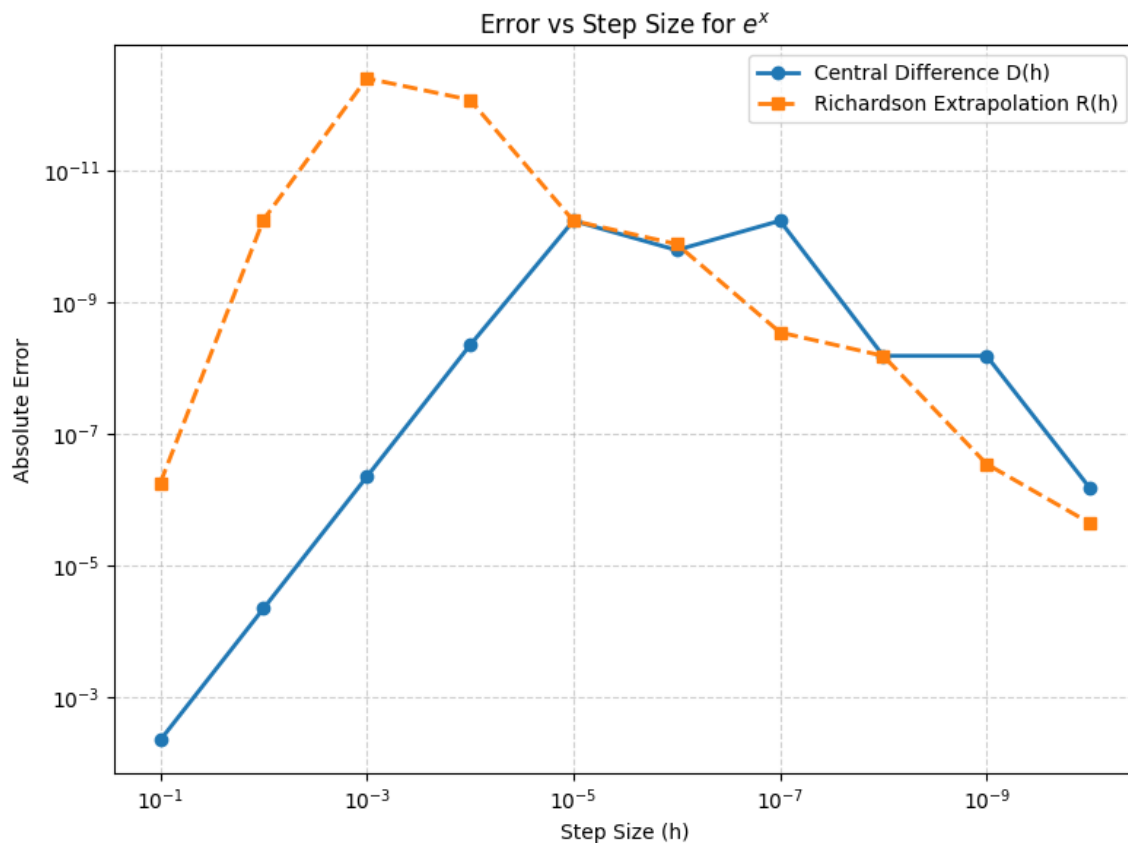


Figure 2: Log-log plots of absolute error versus step size h function (e^x)

Function: $f(x) = \sin(x)$ (at $x = 1$)

h	Exact	D_h	Error_D	R_h	Error_R
0.1	0.540302	0.539402	0.000900054	0.540302	1.12529e-07
0.01	0.540302	0.540293	9.00499e-06	0.540302	1.1276e-11
0.001	0.540302	0.540302	9.00505e-08	0.540302	3.54161e-14
0.0001	0.540302	0.540302	9.0043e-10	0.540302	1.07159e-12
1e-05	0.540302	0.540302	1.11409e-11	0.540302	1.10636e-11
1e-06	0.540302	0.540302	2.77169e-11	0.540302	2.77169e-11
1e-07	0.540302	0.540302	1.94328e-10	0.540302	9.34476e-10
1e-08	0.540302	0.540302	2.58123e-09	0.540302	4.82026e-09

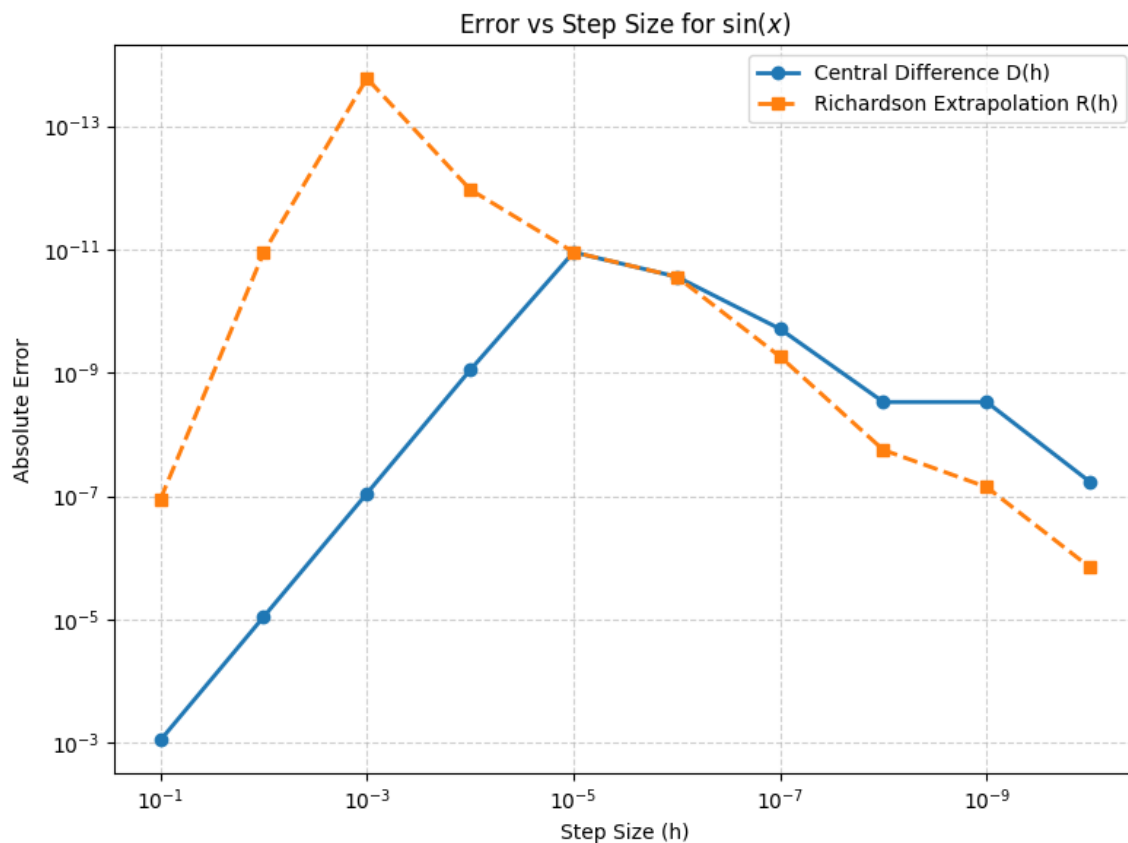


Figure 3: Log-log plots of absolute error versus step size h function ($\sin(x)$)

Function: $f(x) = \cos(x)$ (at $x = 1$)

h	Exact	D_h	Error_D	R_h	Error_R
0.1	-0.841471	-0.840069	0.00140175	-0.841471	1.75254e-07
0.01	-0.841471	-0.841457	1.40244e-05	-0.841471	1.75419e-11
0.001	-0.841471	-0.841471	1.40245e-07	-0.841471	1.13354e-13
0.0001	-0.841471	-0.841471	1.40253e-09	-0.841471	1.53355e-12
1e-05	-0.841471	-0.841471	1.08641e-11	-0.841471	3.46267e-12
1e-06	-0.841471	-0.841471	2.75173e-11	-0.841471	1.20512e-10
1e-07	-0.841471	-0.841471	3.0555e-10	-0.841471	1.17475e-09
1e-08	-0.841471	-0.841471	3.02512e-09	-0.841471	1.78281e-08

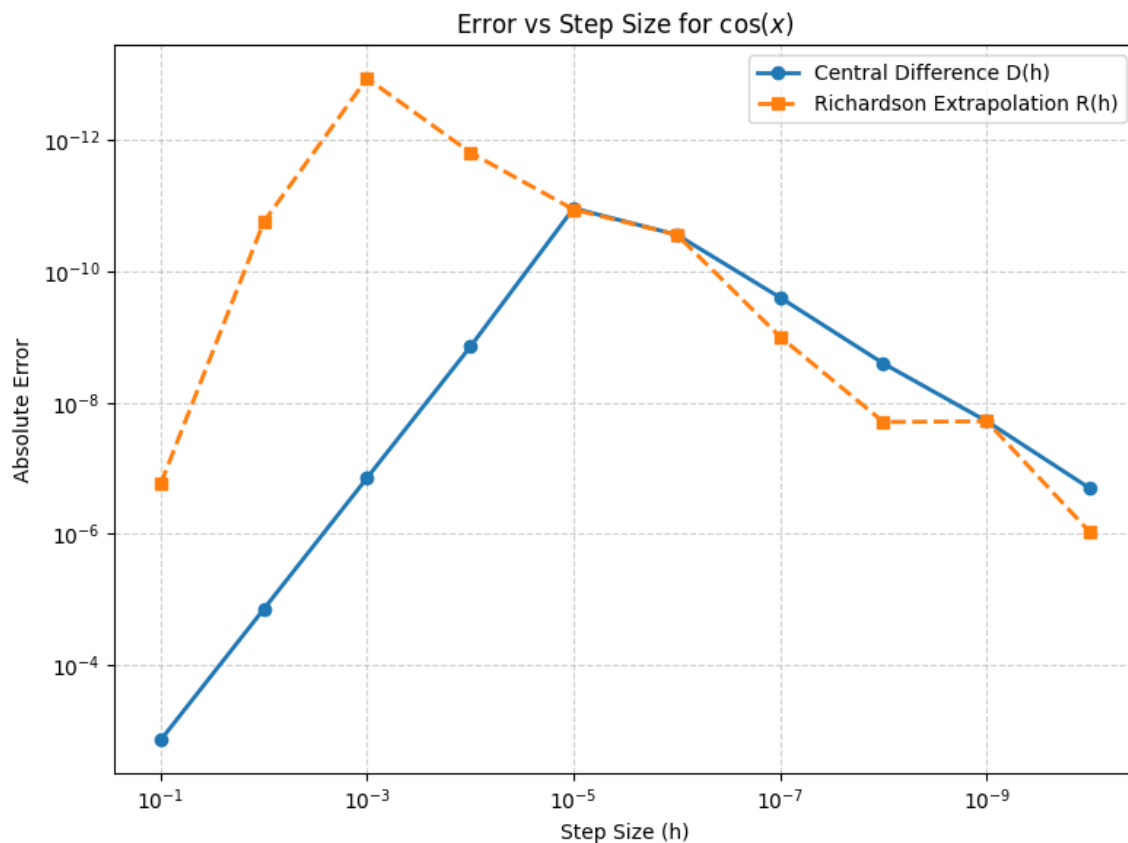


Figure 3: Log-log plots of absolute error versus step size h function ($\cos(x)$)

Function: $f(x) = x^3 - 2x + 1$ (at $x = 1$)

h	Exact	D_h	Error_D	R_h	Error_R
0.1	1	1.01	0.01	1	1.9984e-15
0.01	1	1.0001	0.0001	1	2.498e-14
0.001	1	1	1e-06	1	7.31637e-14
0.0001	1	1	9.99967e-09	1	2.60014e-13
1e-05	1	1	9.5369e-11	1	3.61571e-11
1e-06	1	1	2.87557e-11	1	1.76785e-10
1e-07	1	1	5.83867e-10	1	5.83867e-10
1e-08	1	1	6.07747e-09	1	6.07747e-09

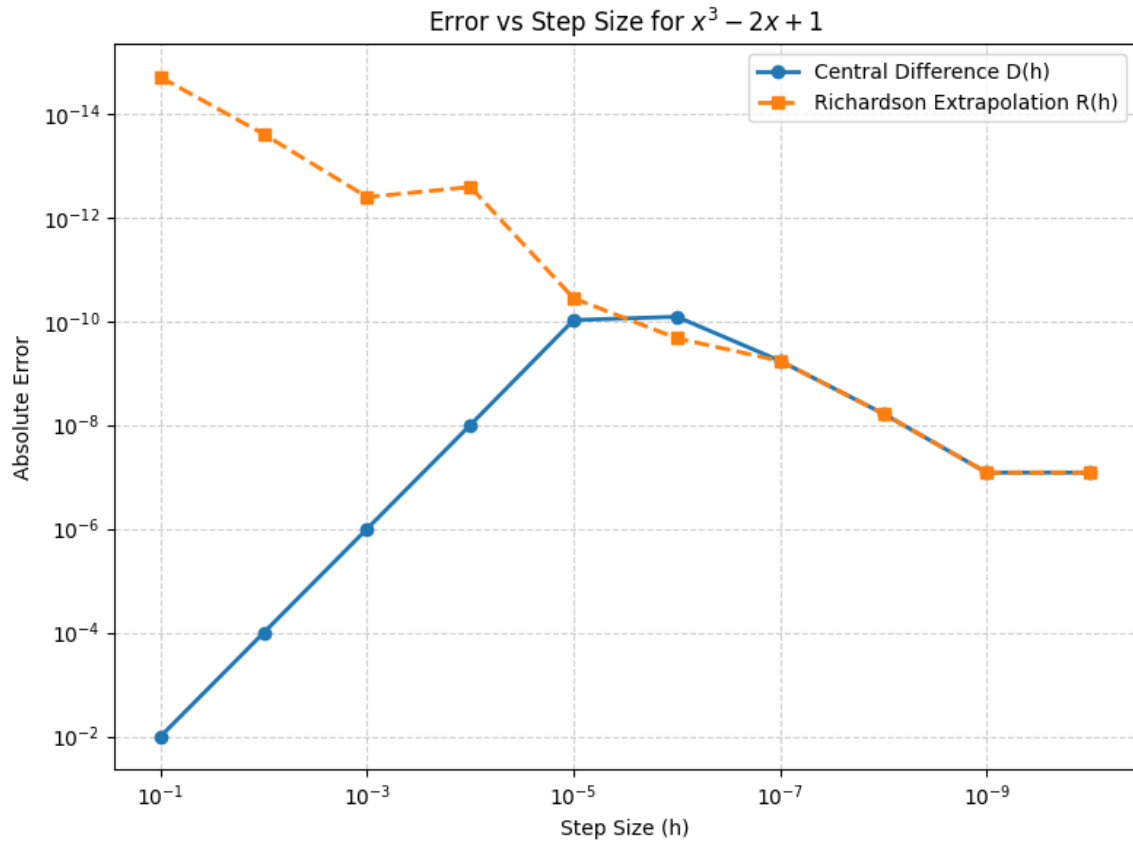


Figure 4: Log-log plots of absolute error versus step size h function ($x^3 - 2x + 1$)

Slope:

We plotted the absolute errors of Central Difference $D(h)$ and Richardson Extrapolation $R(h)$ against the step size h on a log-log scale using the calculated values from output.txt.

We then examined the approximately linear region of the curves and calculated its slope using $\log(E)$ vs. $\log(h)$ to compare the observed convergence order with the theoretical $O(h^2)$ and $O(h^4)$ behavior.

Observed Central Difference Slope: 2.00
Observed Richardson Slope: 4.00

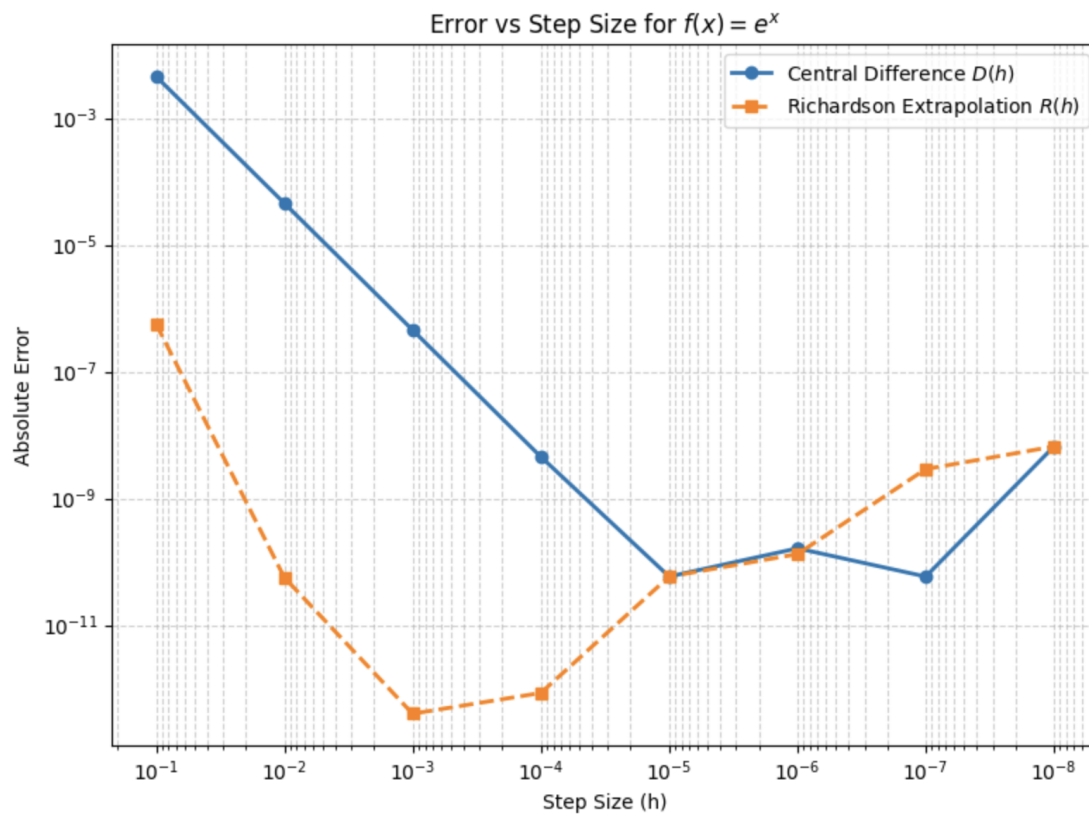


Figure 5: Log-log plots of absolute error versus step size h function (e^x) with grid lines to calculate slope.

Answers to Assignment Questions

Q1. Does Richardson always give a smaller error than Central Difference?

Yes, in most cases. As long as the step size is moderately small. Richardson extrapolation cleans up the leading error left behind by Central Difference so we get a much closer answer.

Q2. By approximately what factor does the error decrease when h is reduced by 10?

It depends on the method. For Central Difference, reducing h by a factor of 10 makes the error go down by about 100 times (10^2). For Richardson Extrapolation, reducing it by a factor of 10 makes the error decrease by around 10,000 times (10^4), before round-off errors dominate.

Q3. Does Richardson continue to improve as h becomes very small?

No. While truncation error decreases as h becomes smaller, floating-point round-off error increases because computer handles only few finite decimal places. Once round-off error begins to dominate, further reducing h causes the error to plateau or increase.

Q4. Does the log-log slope support the theoretical prediction?

Yes, it matches well. When we plot the data on a log-log graph, it forms a straight line where the slope represents the order of accuracy. The observed slopes align closely with the theoretical predictions of 2 for Central Difference and 4 for Richardson Extrapolation in the valid region.

Q5. Does Richardson achieve the expected $O(h^4)$ behavior?

Yes, the data confirms that Richardson extrapolation achieves $O(h^4)$ convergence, shown by the rapid drop in error values reaching near machine precision limits before round-off begins.

Q6. What role does floating-point round-off error play?

It acts as a lower bound on achievable accuracy. In numerical differentiation, subtracting very close function values in the numerator leads to minute cancellation of a very tiny 0.000000000000001 when h is extremely small. This causes the total error curve to curve upward at very small values of h .